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Fall 2019-2020 Q1:

a) $y' = e^x \sec(y) + \sec(y)$ $\boxed{\times \cos(y)}$

$\cos(y)\frac{dy}{dx} = e^x + 1 \rightarrow \int \cos(y)\,dy = \int e^x + 1\,dx$

$\sin y = e^x + x + C.$

b) $\frac{dy}{dx} = -\sin(x-y) + 1$ $\quad z = x - y \rightarrow \frac{dz}{dx} = 1 - \frac{dy}{dx}$

$\cancel{1} - \frac{dz}{dx} = -\sin z + \cancel{1}$

$\int \frac{1}{\sin z} dz = \int dx \rightarrow \int \csc z\, dz = \int dx$

$\ln|\csc z - \cot z| = x + C.$

$\hookrightarrow \ln|\csc(x-y) - \cot(x-y)| = x + C.$

c) $(3x^2 + 2y\sin(2x))dx + (2\sin^2(x) + 3y^2)\,dy = 0$

$\frac{\partial M}{\partial y} = ~~[illegible]~~ 2\sin 2x \qquad \frac{\partial N}{\partial x} = 2\sin 2x \qquad \therefore$ Exact

$\int M\,dx + \int N\,dy = C$

$x^3 + 4y\cos 2x + y - y\cos 2x + y^3 = C$

$x^3 + y + y^3 + 3y\cos 2x = C.$

d) $(1+x^2)\frac{dy}{dx} - 2xy = x\,y^3 \quad \div (1+x^2)$

$y' - \frac{2x}{1+x^2} = \frac{x}{1+x^2} y^3 \qquad \mu = e^{-\int \frac{2x}{1+x^2}(1-3)} = e^{2\ln|1+x^2|} = e^{\ln(1+x^2)^2} = (1+x^2)^2$

$y = \frac{1}{\mu}\left[\int \mu \cdot q(x)(1-n)\,dx + c\right] = (1+x^2)^{-2}\left[-\int (1+x^2)^{2} \cdot \frac{x}{\cancel{1+x^2}} \cdot \frac{(-2)}{-1}\right]$

$= (1+x^2)^{-2}\left[-\frac{(1+x^2)^2}{2} + C.\right]$

$= \frac{-1}{2} + (1+x^2)^{-2} C.$

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Q2:7

ⓐ $y^5 - 3y^4 + 6y^3 - 4y^2 = 0$ $m^5 - 3m^4 + 6m^3 - 4m^2 = 0$

$m^2(m^3 - 3m^2 + 6m - 4) = 0$

$m_{1,2} = 0$ $m_3 = 1$ $m_{4,5} = 1 \pm \sqrt{3}i$

$y = C_1 e^{0x} + C_2 x e^{0x} + C_3 e^x + e^x [C_4 \cos\sqrt{3}x + C_5 \sin\sqrt{3}x]$.

b) $y'' + y' + y = 2\sin(x)$

$m^2 + m + 1 = 0$ $u = \cos\frac{\sqrt{3}}{2}x$ $v = \sin\frac{\sqrt{3}}{2}x$

$m_{1,2} = \frac{-1}{2} \pm \frac{\sqrt{3}}{2}i$

$y_{c.F} = e^{-\frac{1}{2}x}[C_1 \cos\frac{\sqrt{3}}{2}x + C_2 \sin\frac{\sqrt{3}}{2}x]$. $u' = -\sin\frac{\sqrt{3}}{2}x$ $v' = \cos\frac{\sqrt{3}}{2}x$

$uv' - vu' = 1$

$A(x) = \int \frac{-v f(x)}{uv' - vu'} = \int -\sin\frac{\sqrt{3}}{2}x \cdot 2\sin x \, dx$

$= \cos\frac{\sqrt{3}}{2}x \cdot -2\cos x + C.$

$B(x) = \int \frac{u f(x)}{uv' - vu'}$ $\boxed{y_{P.I} = A(x) + B(x)}$

$= \int \cos\frac{\sqrt{3}}{2}x \cdot 2\sin x = \sin\frac{\sqrt{3}}{2}x \cdot -2\cos 2x + k.$

$y_{g.s} = y_{c.F} + y_{P.I}$

c) $y'' + y = \tan(x)\sec(x)$ $m^2 + 1 = 0$ $m_{1,2} = \pm i$

$u = \cos x$ $v = \sin x$ $y_{c.f} = e^{0x}[C_1 \cos x + C_2 \sin x]$

$u' = -\sin x$ $v' = \cos x$

$uv' - vu' = 1$

$A(x) = \int \frac{-v f(x)}{uv' - vu'} = \int -\sin x \cdot \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} = \int \frac{-\sin^2 x}{\cos^2 x} = \int \frac{-1 + \cos^2 x}{\cos^2 x}$

$= \int \frac{-1}{\cos^2 x} + 1 = \int \sec^2 x + 1 = \tan x + x + C.$

$B(x) = \int \frac{u f(x)}{uv' - vu'} = \int \cancel{\cos x} \cdot \frac{\sin x}{\cancel{\cos x}} \cdot \frac{1}{\cos x} = \int \tan x = \ln|\sec x| + k.$

$y_{g.s} = e^{0x}[C_1 \cos x + C_2 \sin x] + \tan x + x + C + \ln|\sec x| + k.$

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d) $y''+y=1$

نبدأ تعويض هنا

من حوله

$y=\cos x$

$y=V\cos x$

$y'=\frac{dv}{dx}+\cos x-V\sin x$

$y''=\frac{d^2v}{dx^2}\cos x-\sin x\frac{dv}{dx}-V\cos x$
$-\sin x\frac{dv}{dx}$

$$\frac{d^2v}{dx^2}\cos x-2\sin x\frac{dv}{dx}-\cancel{V\cos x}\cancel{+V\cos x}=1$$

$$\frac{dz}{dx}\cos x-2\sin x\,Z=1 \quad \boxed{\div\cos x}$$

$\frac{dv}{dx}=Z$

$\frac{d^2v}{dx^2}=\frac{dz}{dx}$

$$\frac{dz}{dx}-2\tan x\,Z=\frac{1}{\cos x}$$

$$\mu=e^{-2\int\tan x\,dx}=e^{-2\ln\sec x}=\cancel{e}^{\cancel{\ln}(\sec x)^{-2}}=(\sec x)^{-2}$$

$$Z=(\sec x)^2\left[\int(\sec x)^{-2}\cdot\frac{1}{\cos x}\,dx+c\right]$$

$$=(\sec x)^2\left[\int\frac{1}{\sec x}\,dx+c\right]=\sec^2 x\left[\sin x+c\right]$$

$$\frac{dv}{dx}=\sec^2 x\left[\sin x+c\right]=\tan x\cdot\sec x+\sec^2 x\,c$$

$$\int dv=\int\tan x\cdot\sec x+\sec^2 x\,c\,dx$$

$$V=\sec x+\tan x\,c+K.$$

$$\frac{y}{\cos x}=\sec x+\tan x+K$$

$$\boxed{y=1+\sin x+K}\ \#$$

Q_3:↓ ⓐ $y'' - 3y' + 2y = 4$ $y(0) = 1$ $y'(0) = 0$

$$S^2 y(S) - S\underset{\to 1}{\cancel{y(0)}} - \cancel{y'(0)} - 3\,S y(S) - 3\,\underset{\to 1}{\cancel{y(0)}} + 2y(S) = \frac{4}{S}$$

$$y(S)\left[S^2 - 3S + 2\right] = \frac{4}{S} + \frac{4}{1} = \frac{4+4S}{S}$$

$$y(S) = \frac{4+4S}{S(S-1)(S-2)} = \frac{A}{S} + \frac{B}{S-1} + \frac{C}{S-2}$$

$\boxed{A = 2}$ $\boxed{B = -8}$ $\boxed{C = 6}$

$$y(S) = \frac{2}{S} + \frac{-8}{S-1} + \frac{6}{S-2}$$

$$y(t) = 2 + (-8)e^{t} + 6e^{2t}.$$

ⓑ ⓘ $f(t) = \cos(3t + \frac{\pi}{2}) + \cosh t$

$$f(s) = \frac{s}{s^2+9} + \frac{s}{s^2+(\frac{\pi}{2})^2} + \frac{s}{s^2-1}.$$

$(t-5)^2 + 10(t-5) + 25$ ↗

هنزود ⑤ وأنقص
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عملت حاجة

ⓘⓘ $f(t) = u_5(t)\, e^{t-5}\, t^2 = u_5(t)\, e^{t-5}\,((t-5)+5)^2 = u_5(t)\, e^{t-5}(t-5)^2 + 10 u_5(t)\, e^{t-5}(t-5) + 25 u_5(t)\, e^{t-5}(1)$

$$f(s) = e^{-5s}\frac{2!}{\cancel{e}S^3} + 10e^{-5s}\frac{1!}{S^2} + 25e^{-5s}\frac{1}{s}$$

ⓘⓘⓘ $f(t) = t\,(\sin t + e^{2t})$

$$F(s) = \frac{-d}{ds}\left(\frac{1}{s^2+1} + \frac{1}{s-2}\right) = \frac{2s}{(s^2+1)^2} + \frac{1}{(s-2)^2}.$$

ⓘⓥ $f(t) = \sin 2t \cos 2t + \int_0^t \cosh\tau + 1\, d\tau$

$$f(t) = \sin 4t - \sin 0 + \int_0^t \cosh\tau + 1\, d\tau$$

$$f(s) = \frac{4}{s^2+16} - \frac{0}{s^2+0} + \frac{1}{s}\left[\frac{s}{s^2-1} + \frac{1}{s}\right].$$

Q4: ⓐ

i) $F(s) = \dfrac{s^3 - s^2 + s - 1}{s^5} + \dfrac{4s}{s^2+16}$

$F(s) = \dfrac{1}{s^2} - \dfrac{1}{s^3} + \dfrac{1}{s^4} - \dfrac{1}{s^5} + 4\dfrac{s}{s^2+16}$.

$F(t) = t - \dfrac{1}{2!}t^2 + \dfrac{1}{3!}t^3 - \dfrac{1}{4!}t^4 + 4\cos 4t$.

ii) $F(s) = \dfrac{e^{-s} - e^{-2s}}{s^2-1} = \dfrac{e^{-s}}{s^2-1} - \dfrac{e^{-2s}}{s^2-1}$.

$f(t) = U(t)\sinh(t-1) - U_2(t)\sinh(t-2)$.

iii) $F(s) = \dfrac{1}{s(s^2+1)}$

$f(t) = \int_0^t \sin\tau \, d\tau = [\cos\tau]_0^t = [\cos t - \cos 0]$.

iv) $F(s) = \ln\left(\dfrac{s+6}{s+2}\right) = \ln(s+6) - \ln(s+2)$

$f'(s) = \dfrac{1}{s+6} - \dfrac{1}{s+2}$ $\quad -t\,f(t) = e^{-6t} - e^{-2t}$

$f(t) = \dfrac{e^{-6t} - e^{-2t}}{-t}$

ⓑ $f(x) = x$ $\quad$ ∵ Fourier Cosine Series

$\therefore a_0 = \text{Zero}$

$f(x) = \dfrac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos \dfrac{n\cancel{\pi}}{\cancel{\pi}}x + b_n \sin \dfrac{n\cancel{\pi}}{\cancel{\pi}}x$

$a_n = \int_{-\pi}^{\pi} x \cos nx \, dx$

$$\begin{array}{c|l} x \; \oplus & \cos nx \\ \hline 1 \; \ominus & -\dfrac{\sin nx}{n} \\ 0 & -\dfrac{\cos nx}{n^2} \end{array}$$

$a_n = \dfrac{-x\cancel{\sin n}x}{n} + \dfrac{\overset{(-1)^n}{\cos nx}}{n^2} = \dfrac{(-1)^n}{n^2}$ $\quad \to \text{Zero}$

$b_n = \int_{-\pi}^{\pi} x \sin nx \, dx$

$$\begin{array}{c|l} x \; \oplus & \sin nx \\ \hline 1 \; \ominus & \dfrac{\cos nx}{n} \\ 0 & -\dfrac{\sin nx}{n^2} \end{array}$$

$b_n = \dfrac{x\cos nx}{n} + \dfrac{\cancel{\sin nx}}{n^2} \to \text{Zero}$

$= \dfrac{x(-1)^n}{n}$

$\therefore f(x) = \sum_{n=1}^{\infty} \dfrac{(-1)^n}{n^2}\cos nx + \dfrac{x(-1)^n}{n}\sin nx$